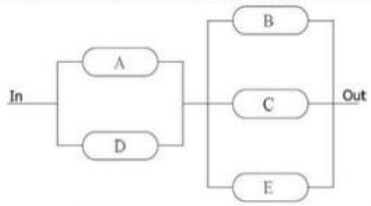
QUIZ 2025/2026

Quiz 1

The system represented in the picture is considered operational if a message going **in** can find a path of properly functional components leading **out**. If a component is not properly functional, the message cannot pass through.



If components A, B, C, D, and E function properly with probabilities $0.73, 0.57, 0.81, 0.84$, and 0.74, respectively, independently of each other.

What is the probability that both components A and E function properly? 0.5402

If component D functions properly, what is the probability that the system fails? 0.0212

Are the two events $F = [A \text{ and } E \text{ fail}]$ and $W = [A \text{ and } E \text{ function properly}]$ independent? (Answer Y for yes or N for no) N

Let A denote the event "Component A works", so $\bar{A}$ corresponds to "Component A fails". Define analogous events for all components. Then

a. By independence, the probability that both components A and E function properly is

$$\mathbb{P}[A \cap E] = \mathbb{P}[A]\,\mathbb{P}[E] = 0.73 \cdot 0.74 = 0.5402.$$

b. If component D functions properly, the probability that the system fails is

$$\mathbb{P}[\bar{C} \cap \bar{D} \cap \bar{E}] = (1 - \mathbb{P}[C])(1 - \mathbb{P}[D])(1 - \mathbb{P}[E]) = (1 - 0.57)(1 - 0.81)(1 - 0.74) = 0.0212$$

c. F and W are mutually exclusive and they both have positive probability, so they cannot be independent.

- 0.5402
- 0.0212
- N

Suppose that a card is extracted at random from a regular deck of 52 cards with 4 suits (diamonds, clubs, hearts and spades) and 13 cards for each suit (numbers from 1 to 10 plus jack, queen and king). Denote the events "picking a heart", "picking a king" and "picking a diamond" by A, B and C respectively. Which of the following statements are true (there may be more than one)?

Select one:

- a. $\mathbb{P}(A|C) = \mathbb{P}(A)$
- b. All elements of the sample space are equiprobable
- c. A and C are independent
- d. A, B and C are mutually exclusive
- e. The collection $\{A, B, C\}$ defines a partition of the sample space

The correct answer is: All elements of the sample space are equiprobable

A online retailer uses 3 different couriers to deliver the orders: courier 1 handles 37% of of the orders, while 36% are handelled by 2 and 27% by 3. It is known from past experience that 4% of the orders handelled by courier 1 are delayed, while for couriers 2 and 3, the percentages of delayed deliveries are 10% and 8%, respectively.

If a delivery is chosen at random, what is the probability that it is delayed? 0.0724

A customer complains for a delayed delivery. What is the probability that it was handelled by courier 3? 0.2983

Let D denote the event "the delivery is delayed" and let C_i denote the event "the delivery was handelled by courier i", $i = 1, 2, 3$.

a. By the law of total probabilitiy

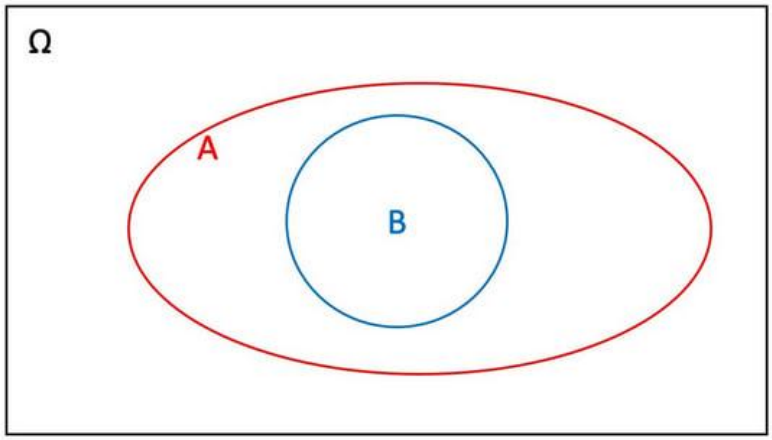
$$\mathbb{P}[D] = \sum_{i=1}^{3} \mathbb{P}[D|C_i]\mathbb{P}[C_i] = 0.0724.$$

b. By Bayes' theorem,

$$\mathbb{P}[C_3|D] = \frac{\mathbb{P}[D|C_3]\mathbb{P}[C_3]}{\mathbb{P}[D]} = 0.2983.$$

- 0.0724
- 0.2983

Consider the two events A and B of the sample space Ω, both with positive probability and represented in the following diagram.



Which of the following is statements is not true?

Select one:

- a. $\mathbb{P}[\bar{A}|B] = \mathbb{P}[\bar{A}]/\mathbb{P}[B]$
- b. $\mathbb{P}[B|\bar{A}] = 0$
- c. $\mathbb{P}[\bar{B}|\bar{A}] = 1$
- d. None. These are all true statements
- e. $\mathbb{P}[B] \leq \mathbb{P}[A]$

The correct answer is: None. These are all true statements

Quiz 2

Consider two discrete random variables, X and Y, such that:

x	$p_X(x)$
0	0.12
1	0.44
2	0.44

	$y=1$	$y=2$
$p_{Y\|X}(y\|0)$	0.4	0.6
$p_{Y\|X}(y\|1)$	0.3	0.7
$p_{Y\|X}(y\|2)$	0.2	0.8

Find $\mathbb{P}[X=1, Y=1]$ 0.132

Find $\mathbb{P}[Y=1]$ 0.268

Find $\mathbb{P}[X<Y]$ 0.428

Are the two variables independent? (Answer Y for yes or N for no) N

a. $\mathbb{P}[X=1, Y=1] = \mathbb{P}[Y=1|X=1]\mathbb{P}[X=1] = 0.3 \cdot 0.44 = 0.132$

b. $\mathbb{P}[Y=1] = \sum_{i=0}^{2} \mathbb{P}[Y=1|X=i]\mathbb{P}[X=i] = 0.268$

c. $\mathbb{P}[X<Y] = \mathbb{P}[X=0, Y=1] + \mathbb{P}[X=0, Y=2] + \mathbb{P}[X=1, Y=2] = 0.428$, where $\mathbb{P}[X=i, Y=j] = \mathbb{P}[Y=j|X=i]\mathbb{P}[X=i]$.

d. The distribution of $Y|X=x$ changes depending on the value of x, therefore the two variables are not independent.

- 0.132
- 0.268
- 0.428
- N

Consider two events, A and B, such that $\mathbb{P}[A] + \mathbb{P}[B] = 1$. Which of the following statements is surely true?

Select one:

- a. $\mathbb{P}[A] = \mathbb{P}[B]$
- b. $\mathbb{P}[A \cup B] = 1$
- c. A and B form a partition of the sample space
- d. $A \cup B = \Omega$
- e. None of the other options is surely true True
- f. $A \cap B = \emptyset$

The correct answer is: None of the other options is surely true

The time, in minutes, it takes to reboot a certain system is a continuous variable with density

$$f(x) = \left[k - \frac{x}{80.5}\right] 1_{(0,9)}(x)$$

. Find k. Important: Use at least 4 decimals and use the "," instead of the "." to denote the decimal point.

Answer: 0.1670

$$1 = \int_0^9 \left(k - \frac{x}{80.5}\right) dx = k9 - \frac{9^2}{2 \cdot 80.5} \quad \Rightarrow \quad k = \frac{1}{9} + \frac{9}{2 \cdot 80.5} = 0.167$$

Notice that

$$0 < 0.0551988 \leq 0.167 - \frac{x}{80.5}, \quad \text{for all } x \in (0, 9)$$

The correct answer is: 0.167

Quiz 3

Unfortunately, the prof. opened the wrong quiz, this should have been the 4th one. Below are the questions.

Hint: Save time by using adequate R functions to calculate the necessary probabilities

Consider two servers. The monthly number of shutdowns of each server follows, independently of the other, a Poisson distribution with mean $\mathbf{0.35}$ for server $\mathbf{1}$ and mean $\mathbf{0.65}$ for server $\mathbf{2}$.

What is the probability that there are at least $\mathbf{2}$ shutdowns to server $\mathbf{1}$ within a year? 0.922023

What is the probability that there are exactly $\mathbf{2}$ shutdowns to server $\mathbf{1}$ and $\mathbf{3}$ shutdowns to server $\mathbf{2}$ within a year? 0.00428615

What is the probability that there are at exactly $\mathbf{5}$ shutdowns in total to both servers within a year? 0.01274064

What is the expected total number of shutdowns to both servers within a year? 12

Within a year, what is the probability that there are at least $\mathbf{2}$ months with no shutdowns of server $\mathbf{2}$? 0.9575587

What is the expected number of months per year with no shutdowns to server $\mathbf{2}$? 6.264549

a. Let X be the yearly number of shutdowns to server $\mathbf{1}$. Then, $X \sim \mathrm{Po}(12 \cdot 0.35)$ and

$$\mathbb{P}[X \geq 2] = 1 - \mathtt{ppois(1, 12 * 0.35)} = 0.922.$$

b. Now, let $Z \sim \mathrm{Po}(12, 0.65)$ be the yearly number of shutdowns to server $\mathbf{2}$
Then,

$$\mathbb{P}[X = 2, Z = 3] = \mathbb{P}[X = 2]\,\mathbb{P}[Z = 3] = \mathtt{dpois(2, 12 * 0.35) * dpois(3, 12 * 0.65)} = 0.0043.$$

(c) Let $S = X + Z$ be the total number of yearly shutdowns.
By the *law of total probability*:

$$\begin{aligned}\mathbb{P}[S = 5] &= \sum_{k=0}^{5} \mathbb{P}[X = k]\,\mathbb{P}[Z = 5 - k] \\ &= \sum_{k=0}^{5} \mathtt{dpois}(k, 12 * 0.35)\,\mathtt{dpois}(5 - k, 12 * 0.65) \\ &= 0.0127.\end{aligned}$$

(d) Using the *linearity of expectation*:

$$\mathbb{E}[S] = \mathbb{E}[X] + \mathbb{E}[Z] = 12\,0.35 + 12\,0.65 = 12.$$

e. Let Y be the number of months with no shutdowns to server $\mathbf{2}$ within a year. Then, $Y \sim \mathrm{Bin}(12, p)$, where

$$p = \mathtt{ppois(0, 0.65)} = 0.522.$$

Finally,

$$\mathbb{P}[Y \geq 2] = 1 - \mathtt{pbinom(1, 12, 0.522)} = 0.998.$$

f. $\mathbb{E}[Y] = 12 \cdot 0.522 = 6.2645.$

- 0.922
- 0.0043
- 0.0127
- 12
- 0.998
- 6.2645

A logistics platform receives a daily random number of orders. The number N of orders in a given day follows a Poisson distribution with parameter $\lambda = 1.4$. Each order is successfully delivered with probability $p = 0.45$, independently of the others. In other words, given $N = n$ the daily number X of successfully delivered orders follows a Binomial distribution, with parameters n and p.

Using the law of total expectation and the law of total variance, compute:

The mean of X 0.63

The variance of X 0.63

a. We know that $\mathbb{E}[N] = \mathrm{Var}[N] = \lambda$. By the law of total expectation,

$$\mathbb{E}[X] = \mathbb{E}(\,\mathbb{E}[X \mid N]\,) = \mathbb{E}[Np] = p\,\mathbb{E}[N] = (0.45)\,(1.4) = 0.63.$$

b. By the law of total variance,

$$\mathrm{Var}(X) = \mathbb{E}[\mathrm{Var}(X \mid N)] + \mathrm{Var}(\mathbb{E}[X \mid N]).$$

Since $X \mid N \sim \mathrm{Bin}(N, p)$,

$$\mathrm{Var}(X \mid N) = Np(1-p), \quad \mathbb{E}[X \mid N] = Np.$$

Therefore,

$$\begin{aligned}\mathrm{Var}(X) &= \mathbb{E}[Np(1-p)] + \mathrm{Var}(Np)\\ &= p(1-p)\,\mathbb{E}[N] + p^2\,\mathrm{Var}(N)\\ &= p(1-p)\,\lambda + p^2\,\lambda\\ &= p\,\lambda = 0.45\,1.4\\ &= 0.63.\end{aligned}$$

- 0.63
- 0.63